

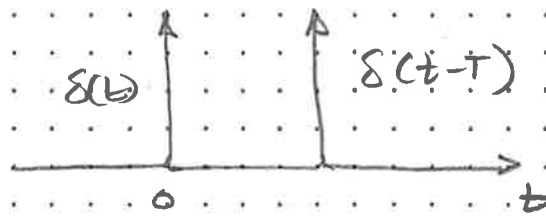
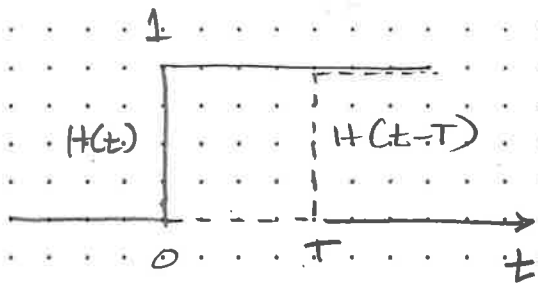
STEP FUNCTION & DELTA FUNCTION

$H(t)$

["JUMP" FUNCTION]

$\delta(t)$

["IMPULSE" FUNCTION
"SLAP" FUNCTION]



$$H(t) = \begin{cases} 0 & ; t < 0 \\ 1 & ; t \geq 0 \end{cases}$$

$$\frac{dH}{dt} = \delta(t)$$

→ [standard jump function]

→ "Jumps" from 0 to 1 at $t=0$

→ "Jump function is constant
i.e., 0 for $t < 0$
& 1 for $t \geq 0$

$H(t-T)$ → [Shifted jump function]

→ [At $t=T$]

$$\begin{aligned} &\rightarrow H(t-T) \\ &= H(T-T) = H(0) \end{aligned}$$

Jump occurs at $H(0)$

So, its derivative

→ at $t < 0$

$$\left[\frac{dH}{dt} = 0 \right]$$

→ at $t > 0$

$$\left[\frac{dH}{dt} = 0 \right]$$

→ at $t=0$

$$\left[\frac{dH}{dt} = \infty = \delta(t) \right]$$

Same for the slope or derivative of the shifted jump function.

$$\left[\begin{array}{l} \text{Derivative of the} \\ \text{jump function} \end{array} \right] = \left[\text{Delta-function} \right]$$

$$\boxed{\frac{dH}{dt} = \delta(t)}$$

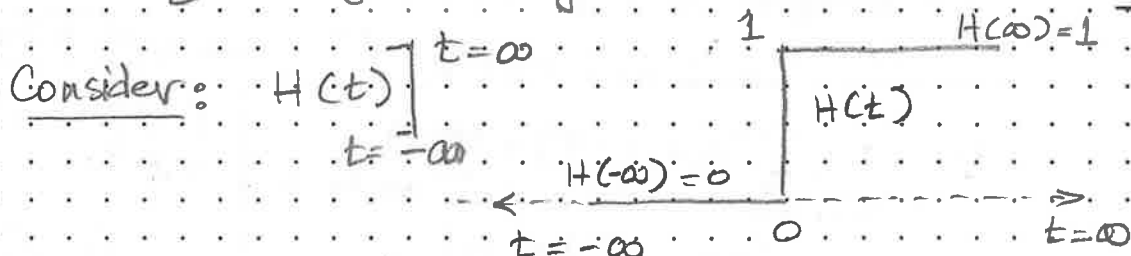
$$\left[\begin{array}{l} \text{Integral of the} \\ \text{Delta function} \end{array} \right] = \left[\text{Jump function} \right]$$

$$\int_{t=-\infty}^{t=\infty} \delta(t) dt = H(t) \Big|_{t=-\infty}^{t=\infty} = 1 - 0 = 1$$

→ The part of calculus that works correctly is Integration.

↳ Tends to make things smoother

→ $\left[\begin{array}{l} \text{Integral of the Delta-function} \\ \text{is a jump function. And, that's} \\ \text{how you really know the } \delta\text{-function} \end{array} \right]$



GENERALIZATION OF DELTA FUNCTION

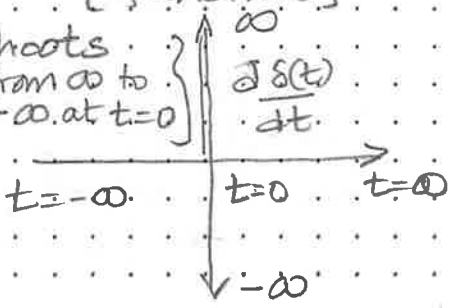
$\delta(t)$

Derivative of $\delta(t)$

$\frac{d\delta(t)}{dt}$

Truly crazy function

shoots from ∞ to $-\infty$ at $t=0$



Integrals of $\delta(t)$

$$\int_{t=-\infty}^{t=\infty} \delta(t) f(t) dt = f(0)$$

$\delta(t)$ is doing everything at $t=0$

$f(t)$ is any other function

At $t=0$, at the same moment, $f(t)$ is doing $f(0)$

EXAMPLES OF INTEGRALS OF FUNCTIONS INVOLVING $\delta(t)$

① $f(t) = 1$ $\int_{-\infty}^{\infty} \delta(t) f(t) dt = f(0)$

$\Rightarrow \int_{-\infty}^{\infty} \delta(t) (1) dt = f(0) = 1$

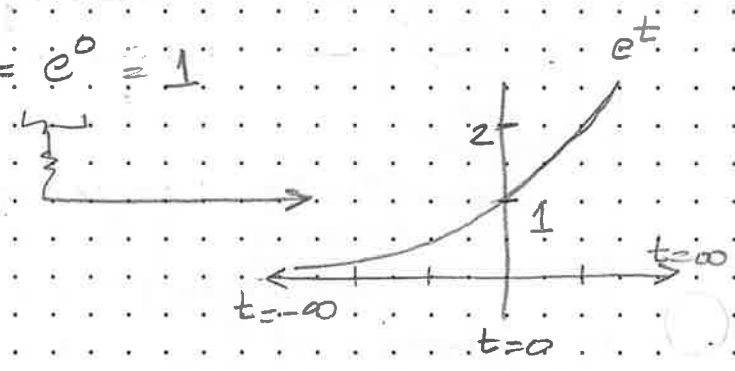
(2) $f(t) = \sin(t)$

$$\int_{-\infty}^{\infty} \delta(t) \sin(t) dt = \sin(0) = 0$$

$\sin(t)$ disappears just when $\delta(t)$ kicks in at $t=0$

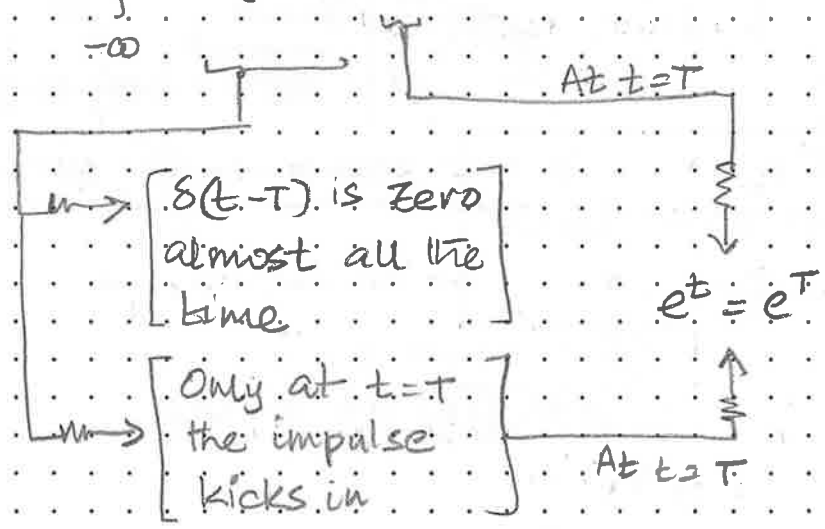
(3) $f(t) = e^t$

$$\int_{-\infty}^{\infty} \delta(t) e^t dt = e^0 = 1$$



(4) $f(t) = e^t$

$$\int_{-\infty}^{\infty} \delta(t-T) e^t dt = e^T \cdot H(t-T) \Big|_{-\infty}^{\infty} = e^T$$



IMPULSE RESPONSE IN FIRST-ORDER LINEAR ODE WITH CONSTANT COEFFICIENTS

If $\delta(t-T)$ was the source term in the first-order linear ODE, what would the solution be?

$$\frac{dy}{dt} = ay + \delta(t-T)$$

δ -function
turned on
at $t=T$

START
WITH
 $y(0)=0$

So, I don't
make any
initial
deposit into
my account

I make one deposit at $t=T$
At that moment, I deposit \$1,
since $\delta(t-T)$ has a unit
coefficient.
If I was depositing \$10, I
would have $10\delta(t-T)$ as my
source term.

Explanation of $\delta(t-T)$ as a source-term, $q(t)$:

$\delta(t-T)$ would say "no input" except at
one moment/instance, T

At that instance, you put in
a deposit!

Remember, $q(t)$ has been a continuous
input, otherwise.

Eg: You put in \$1 per year,
over the whole year!

(91)

KV.B

$\hookrightarrow \delta(t-T) \rightarrow$ says put in \$1 at one moment!

But that's how we do deposits

Now, to the solution of $\left[\frac{dy}{dt} = ay + \delta(t-T) \right]$

with $[y(0)=0]$

$$y(t) = \begin{cases} 0 & ; t < T \\ e^{a(t-T)} & ; t \geq T \end{cases}$$

@ $t < T$ $\left[\begin{array}{l} \text{With no initial deposit } y(0)=0 \\ \& \delta(t-T)=0 \\ y(t)=0 \end{array} \right]$

@ $t = T$ $e^{a(t-T)} = e^{a(T-T)} = 1 = y(T)$

That's our dollar;
that just got
deposited!

@ $t > T$ or $(t-T) > 0$

$$e^{a(t-T)}$$

Growth rate
upto time, t
starting at
earlier time T ,
when the
deposit was made.

Say! $(t-T)=1$
or a year
after T ,

$$y(t) = e^a$$

i.e., a year after
\$1 deposit, my money is
\$ e^a !

FORMAL SOLUTION TO SHIFTED IMPULSE RESPONSE FIRST ORDER LINEAR ODE

(92)
KVB

$$\left[\frac{dy}{dt} = ay + \delta(t-T) \right] \text{ with } [y(0)=0] \text{ as the initial state}$$

$$\frac{dy}{dt} - ay = \delta(t-T)$$

$$\boxed{e^{a(t)} dt} \rightarrow \text{INTEGRATING FACTOR}$$

Consider: $\int a(t) dt = \int -a dt = -at + C$

$$\Rightarrow e^{\int a(t) dt} = ce^{-at} \Rightarrow \text{INTEGRATING FACTOR}$$

$$\cancel{e^{-at}} \frac{dy}{dt} - \cancel{e^{-at}} a y = \cancel{e^{-at}} \delta(t-T)$$

$$\frac{d}{dt}(e^{-at} y) = e^{-at} \delta(t-T)$$

$$d(e^{-at} y) = \delta(t-T) e^{-at} dt$$

$$\int_0^t d(e^{-at} y) = \int_0^t \delta(\tau-T) e^{-a\tau} d\tau$$

$$\cancel{e^{-at}} y(t) - \cancel{e^{-a(0)}} y(0) = \int_0^t \delta(\tau-T) e^{-a\tau} d\tau$$

$$\left[\begin{array}{l} y(0)=0 \\ \text{initial state} \end{array} \right]$$

$$\therefore e^{-at} y(t) = \int_0^t \delta(\tau-T) e^{-a\tau} d\tau$$

Here, The upper limit of the integral is a variable. "T" is a dummy variable

$$= \begin{cases} 0 & ; t < T \\ e^{-aT} & ; t \geq T \end{cases}$$

$$\therefore y(t) = \begin{cases} 0 & ; t < T \\ e^{a(t-T)} & ; t \geq T \end{cases}$$